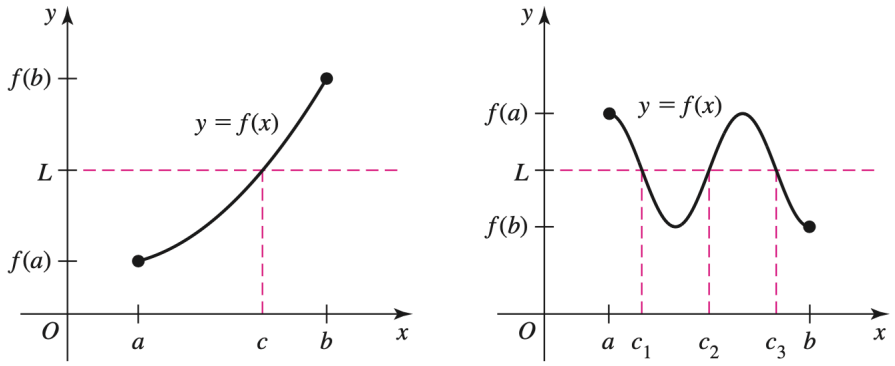
1 Intermediate Value Theorem

Theorem 1.1 (Intermediate Value Theorem (IVT))**.** *If f is continuous on $[a,b]$, then for any real number L between $f(a)$ and $f(b)$ there exists at least one $c \in [a,b]$ such that $f(c) = L$. In other words, if f is continuous on $[a,b]$, then the graph must cross the horizontal line $y = L$ at least once between the vertical lines $x = a$ and $x = b$.*



We now explore the implications of the IVT, postponing the proof until later.

Example 1.2. Suppose we want to know whether the equation $x^3 + x + 1$ has a solution on $[-1,1]$. Let $f(x) = x^3 + x + 1$. Since $f(-1) = (-1)^3 - 1 + 1 = -1$ and $f(1) = 1^3 + 1 + 1 = 3$, by the IVT there exists $c \in [-1,1]$ such that $f(c) = 0$. This proves that $x^3 + x + 1 = 0$ has at least one solution on $[-1,1]$.

1. Does the equation $\cos(x) - \sin(x) - x = 2$ have a solution on the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

2. Is there a number that is exactly 1 more than its cube?

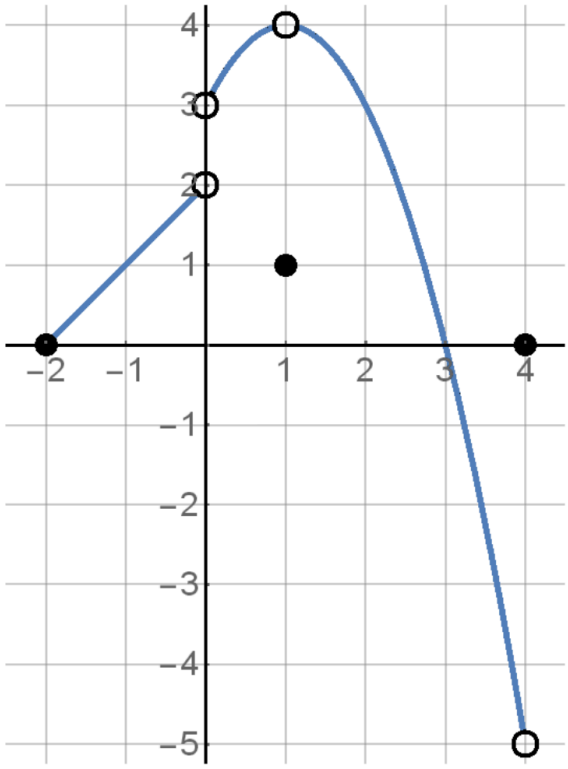
3. What, if anything, does the IVT say about the following? Explain.

   (a) A week ago, there were 23 people in class, but now there are 21 people.

   (b) I was 23 miles from home two hours ago, and now I am 2 miles from home.

4. Let $f(x) = \frac{|x|}{x}$. Then $f(-2) = -1$ and $f(2) = 1$. Therefore, $f(-2) < 0 < f(2)$, but there is no value of c between -2 and 2 for which $f(c) = 0$. Does this violate the IVT? Explain.

5. Let f be the function graphed on the interval $[-2,4]$ as shown to the right.



(a) Where is f discontinuous on the interval $[-2,4]$?

(b) Are the hypotheses of the IVT true on $[-2,3.5]$? Explain.

(c) Is the conclusion of the IVT true on $[-2,3.5]$? Explain.

(d) Are the hypotheses of the IVT true on $[-0.5,0.5]$? Explain.

(e) Is the conclusion of the IVT true on $[-0.5,0.5]$? Explain.

(f) If the hypotheses of the IVT are false, is the conclusion necessarily false?

6. Suppose f is continuous on $[a,b]$ where $f(a) > 0$ and $f(b) > 0$. Can we conclude that $f(x)$ is never zero on $[a,b]$? (*Hint*: Consider $f(x) = \cos(x)$ on $[0, 2\pi]$.)

7. Use the IVT to prove the following fixed point theorem: "If f is continuous on $[0,1]$ and has a range contained in $[0,1]$, then f has a fixed point on $[0,1]$; that is, there is an input $x \in [0,1]$ such that $f(x) = x$. (*Hint*: Consider $g(x) = f(x) - x$.)

8. A Tibetan monk leaves the monastery at 7:00AM and takes his usual path to the top of the mountain, arriving at 7:00PM. The following morning, he starts at 7:00AM at the top and takes the same path back, arriving at the monastery at 7:00PM. Prove that there is a point on the path that the monk will cross at exactly the same time of day on both days. (*Hint*: Let $f(t)$ be the monk's distance from the monastery at time t, where $t = 0$ is 7:00AM on the first day, and let $g(t)$ be his distance from the monastery at time t, where $t = 0$ is 7:00AM on the second day.)